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Are hybrid sit-stand postures a good compromise between sitting and standing?

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Abstract

Potential alternatives for conventional sitting and standing postures are hybrid sit-stand postures (i.e., perching). The purposes of this study were i) to identify where lumbopelvic and pelvic angles deviate from sitting and standing and ii) to use these breakpoints to define three distinct postural phases: sitting, perching, and standing, in order to examine differences in muscle activations and ground reaction forces between phases. Twenty-four participants completed 19 one-minute static trials, from sitting (90°) to standing (180°), sequentially in 5°trunk-thigh angle increments. The perching phase was determined to be 145–175° for males and 160–175° for females. For both sexes, knee extensor activity was lower in standing compared to perching or sitting ($p<0.01$). Anterior-posterior forces were the highest in perching ($p<0.001$), requiring approximately 15% of body-weight. Chair designs aimed at reducing the lower limb demands within 115–170° trunk-thigh angle may improve the feasibility of sustaining the perched posture.

Keywords: ergonomics, low back pain, lumbopelvic angles, perching, chair design

Practitioner Summary

Individuals who develop low back pain in sitting or standing may benefit from hybrid sit-stand postures (perching), yet kinematic and kinetic changes associated with these postures have not been investigated. Perching can improve lumbar posture at a cost of increased lower limb demands, suggesting potential avenues for chair design improvement.

1. Introduction

Standing (Nelson-Wong and Callaghan 2010a; Marshall, Patel, and Callaghan 2011; Gallagher, Campbell, and Callaghan 2014) and sitting (Schinkel-Ivy, Nairn, and Drake 2013; De Carvalho and Callaghan 2011) postures, when maintained statically for a prolonged period of time, are linked to the development of low back pain (LBP) and discomfort (Callaghan et al. 2015). Although the precise cause of this pain and discomfort has not been established, reduced frequency of postural adjustments is consistently linked to the development of LBP in both sitting and standing. It has been shown that postural adjustments can relieve discomfort for a short period of time in studies where taking breaks from sitting or standing to perform different activities. For example, alternating sitting and standing (Gallagher, Campbell, and Callaghan 2014; Karakolis, Barrett, and Callaghan 2016), standing on a sloped surface (Gallagher, Wong, and Callaghan 2013; Gallagher and Callaghan 2016; Nelson-Wong and Callaghan 2010b), raising one leg on a footrest (Gallagher 2014), and walking (De Carvalho 2015), led to a significant reduction in low back discomfort for a short period of time. When standing without the freedom to move around or sit, a higher prevalence of LBP is reported (Tissot, Messing, and Stock 2009). Further, subclinical individuals who develop LBP in standing tend to use less lumbar flexion/extension fidgets and body weight transfers in the first 15 minutes of standing compared to individuals who do not develop LBP (Gallagher and Callaghan 2015). Sitting tends to impose the greatest restriction on postural movements, as the frequency of medial/lateral (M/L) and anterior/posterior (A/P) shifts in sitting is less than half that observed in standing or perching (Le and Marras 2016), indicating that subclinical pain developers in particular may benefit from having an option to assume hybrid sit-stand postures (i.e., perching).

Although perching encompasses a range of postures from sitting to standing, limited empirical evidence exists to suggest the relationship between trunk-thigh angle and lumbar spine orientation. Sitting and standing, two postures often assumed by office workers, lie at the opposite ends of the spectrum in terms of commonly adopted sagittal spine angles: sitting can put joints of the lumbar spine close to the end range of flexion (O'Sullivan et al. 2012; Dunk and Callaghan 2005; Beach et al. 2005; Beach et al. 2008; De Carvalho and Callaghan 2011; Alexander et al. 2007) and standing can position the lumbar spine close to the end range of extension (Alexander et al. 2007; Gallagher, Sehl, and Callaghan 2016; Gallagher, Campbell, and Callaghan 2014). Lumbar flexion increases stress on the passive tissues, including the intervertebral discs (Rohlmann et al. 2011) and posterior longitudinal ligaments, which may cause increased reflexive muscle spasm (Shin, D'Souza, and Liu 2009; Solomonow et al. 2003), inflammatory response (Solomonow et al. 2003; Chan, Ferguson, and Gantenbein-Ritter 2011), and the perception of pain. In contrast, lumbar extension may increase facet joint contact (Adams and Hutton 1980), potentially leading to capsule stretch and the perception of pain. Therefore, postures in between these two endpoints may result in spine orientations that relieve pain and discomfort. It has been suggested, based on a small number of participants, that the lumbar angle closest to neutral, with the thigh muscles in passive equilibrium, occurs when the trunk-thigh angle is 135° in the lateral recumbent position (Keegan 1953). Similarly, reclining the chair backrest from 90° to 130° has demonstrated a reduction in lumbar load by approximately 60% (Rohlmann et al. 2011). However, there has yet to be a quantitative study examining a wider range of these hybrid sit-stand postures to determine whether these trunk-thigh angles are in fact optimal in terms of lumbopelvic and pelvic angles.

Sit-stand stools allow users to work near their standing height while offloading some body weight onto the stool to relieve lower limb demands. Compared to sitting, perching can reduce A/P shear force at the low back and encourage users to move more frequently (Le and Marras 2016); however, it is also associated with increased lower leg swelling, decreased hip comfort (Chester, Rys, and Konz 2002), and no difference in low back discomfort (Le and Marras 2016). Compared to standing, perching also reduces A/P shear force at the low back (Le and Marras 2016), though lower limb swelling and discomfort may be higher or lower depending on the footrest position (Le and Marras 2016; Chester, Rys, and Konz 2002; Antle, Vezina, and Cote 2015) and back extensor activity and co-contraction of erector spinae and external oblique are higher in perching (Antle, Vezina, and Cote 2015). As these results show, there are some benefits to the use of sit-stand stools; however, questions remain as to whether perched postures can improve the potentially negative consequences associated with sitting and standing.

As shown in studies above, perching may be a good alternative posture to sitting or standing; however, it also encompasses a large range of postures that have had little research evaluating the biophysical responses. To the author's knowledge, there is no empirical evidence to support which trunk-thigh angle is considered a different phase (i.e., perching) from sitting or standing. Therefore, the primary purpose of this study was to identify where lumbopelvic and pelvic angles in perching deviate significantly from sitting and standing. The first hypothesis tested if there were any differences between males and females in sitting, standing, and postures in between, since sex differences in sitting (e.g. kinematics (Dunk and Callaghan 2005; Beach et al. 2005) and sitting behavior (Le and Marras 2016; Rohlmann et al. 2014)) have been reported in previous studies. The subsequent hypotheses tested whether the hybrid postures were different from sitting or standing. This test was performed for each sex separately and therefore, the

breakpoints from sitting and standing were established for each sex. The secondary purpose was to use these breakpoints to determine three phases: sitting, perching, and standing, and to examine the differences in physical demand (trunk and knee extensor muscle activities and ground reaction forces) between phases. It should be acknowledged that perching, in this study, is defined by a range of trunk-thigh angles, where the lumbopelvic angle is significantly different from sitting or standing.

2. Methods

2.1 Participants

Twenty-four participants, 12 males (age 25.7 ± 2.3 years; height 177.5 ± 5.6 cm; mass 80.2 ± 15.3 kg; BMI 25.4 ± 4.3 kg/m²) and 12 females (age 24.3 ± 2.2 years; height 167.5 ± 6.1 cm; mass 66.0 ± 8.6 kg; BMI 23.5 ± 2.2 kg/m²), with no history of LBP or severe lower limb injuries were recruited for the study. The study protocol was approved by the University Ethics Review Committee, and all participants provided informed consent prior to participation.

2.2 Instrumentation

To characterize the range of postures between sitting and standing, muscle activation levels, kinematics, and ground reaction forces were collected. All three data sets were collected simultaneously using a 16-bit A/D system (Optotrak Data Acquisition Unit II, Northern Digital Inc., Waterloo, ON, Canada).

Electromyography (EMG) data from three muscles - thoracic erector spinae (TES), lumbar erector spinae (LES), and vastus medialis obliquus (VMO) - were collected bilaterally

using pairs of disposable Ag-AgCl surface electrodes (Blue Sensor, Medicotest Inc., Ølstykke, Denmark). TES and LES EMG signals were collected from the skin surface approximately 2.5 cm bilateral to the spinous process of T9 and L3, respectively, over the erector muscles. VMO EMG signals were collected from approximately 5 cm away from the patella and on the line between the anterior superior iliac spine and the joint space anterior to the medial ligament (SENIAM; surface EMG for a non-invasive assessment of muscles). Each pair of electrodes was placed with a 2 cm centre-to-centre inter-electrode distance, along the fibres of the underlying muscle. A reference electrode was placed on the 12th rib, on the ventral-lateral surface of the body. Raw EMG signals were bandpass filtered (10-1000 Hz) and differentially amplified (common-mode rejection ratio: 115 dB at 60 Hz; input impedance: 10 GΩ) (model AMT-8; Bortec, Calgary, AB, Canada). The amplified signals were then collected at a sample rate of 2000 Hz, with a range of ± 5 V. To facilitate comparisons in muscle activity between participants, each muscle's activity level was normalized to its peak obtained through isometric maximum voluntary contractions (MVCs). For the LES and TES muscles, individuals performed a back extension MVC while lying prone with legs held down, upper body hanging over the edge of a massage table, and arms folded across the chest. For the VMO muscles, individuals performed a leg extension MVC while sitting and keeping the knee angle at approximately 120°. For each MVC, participants were asked to ramp up to a maximum effort and hold for three seconds. One MVC trial was performed for each muscle group.

Three-dimensional kinematic data were collected at 50 Hz using an optoelectronic motion capture system (Optotrak Certus, Northern Digital Inc., Waterloo, ON, Canada). Rigid bodies containing at least three markers were placed on the chest, L1, S1, and bilaterally on the lateral side of thighs and shanks, in order to track eight body segments: trunk, lumbar spine,

pelvis, and bilateral thighs and shanks, respectively. Anatomical landmarks defining the proximal and distal endpoints of each segment were digitized and tracked with respect to the rigid bodies: left and right acromion and iliac crest for the trunk segment; left and right 12th rib and iliac crest for the lumbar spine segment; left and right anterior superior iliac spine (ASIS) and posterior superior iliac spine (PSIS) for the pelvis segment, functional hip joint and medial and lateral condyle for each thigh segment; and medial and lateral condyle and malleolus for each shank segment. From these virtual and physical marker data, body segments were created and their orientation was calculated using Visual 3D software (C-Motion, Inc., Germantown, MD, USA). In addition to these segments, kinematic data of the custom-made seating device were also collected to identify the location and orientation of the seat pan. To normalize postures between participants, quiet standing and maximum standing and sitting flexion and extension trials were collected. Each calibration posture was held and collected for five seconds.

Three separate force plates (all manufactured by Advanced Medical Technology, Inc., Watertown, MA, USA) were used to measure ground reaction forces under the left foot (Model MC12), right foot (Model OR6-7) and the custom-made seating device (Model BP900). Force data were collected at 400 Hz.

2.3 Study Protocol

Each participant completed 19 one-minute trials, from sitting (90° trunk-thigh angle) to standing (180° trunk-thigh angle) sequentially in 5° trunk-thigh angle increments, with two minutes of standing rest in between each trial. The custom-made seating device consisted of a medium density fibreboard (MDF) seat pan, without backrest, that freely rotated and translated horizontally and vertically (Figure 1). The location of the seat was standardized such that the

trunk-thigh angle would increase by 5° in each subsequent trial. Participants were not explicitly positioned to desired trunk-thigh angles since instructing the participants to adjust the trunk position can influence lumbar and pelvic angles and muscle activities. To determine 17 hybrid positions, polar coordinates of the seat pan (separated by 5°) were calculated using an ellipse equation with two standardized seat locations as inputs: 1) seat pan location in sitting with ankles, knees, and hips at 90° flexion; and 2) seat pan location in standing with ankles at 90° flexion and knees and hips at 180° (seat pan was moved up/forward and rotated) (Figure 1). Seat pan position in sitting was standardized by aligning the greater trochanter to the rotating hinge of the seat pan. Seat pan position in standing was standardized by aligning the anterior edge of the seat pan with a position on the leg marked during sitting. This method allowed for consistent location of the hip joint with respect to the rotating hinge of the chair and consistent area of contact between the seat pan and buttock/thigh region throughout the trials and between participants.

Figure 1: Sitting (top) and standing (bottom) calibration trials to determine seat positions for the hybrid sit-stand postures. Note that the anterior edge of the seat pan is aligned to the blue mark on the participant's thigh in both calibration postures. The location of the monitor (height and reach), shown on the right side of both images, is adjusted according to the location of each participant's eye level and followed CSA guideline recommendations (CSA-Z412, 2000).

In each trial, participants were instructed to perform a standardized reading task on a standard computer monitor (24" ViewSonic VA2451m-LED; resolution 1920 x 1080) in front of them while keeping their feet flat on the ground and assuming a still, relaxed posture with their hands resting on their thighs. Participants were refrained from offloading their upper body mass through the use of hands on their thighs and instructed to look straight forward while reading. The vertical orientation of the shank segment was monitored in real-time and controlled within $\pm 2^\circ$ of its orientation of the quiet standing calibration posture to ensure postural changes occurred about the knee and hip only as the seating device was moved. Each posture was collected for one minute to avoid accumulation of fatigue associated with maintaining hybrid sit-stand postures. The computer monitor was adjusted for each trial (i.e., throughout the transition) and for each participant based on the CSA ergonomic guideline (CSA-Z412, 2000), so that the top of the monitor was at eye level vertically (view angle should be within approximately 30° from the horizontal plane). The monitor reach distance and the font size were adjusted so that participants could comfortably read the text without leaning forward. The reading task was performed at a self-selected pace, and the reading material on the screen was scrolled down by the experimenter at the request of a participant.

2.4 Data Analysis

All raw EMG data were notch filtered to remove 60 Hz electrical noise and TES and LES EMG data were high pass filtered at 30 Hz to remove cardiac artifact (Drake and Callaghan 2006) using a Butterworth filter (2nd order, dual pass). These signals were then full-wave rectified and low pass filtered at 2.5 Hz using a Butterworth filter (2nd order, dual pass) and normalized to the peak amplitude recorded during the MVC trials (%MVC).

Raw marker data were low pass filtered at 3 Hz to remove high frequency noise using a Butterworth filter (2nd order, dual pass). The Cardan sequence of sagittal-transverse-axial planes was used to calculate the following angles: lumbopelvis (lumbar-pelvis), pelvis, and seat pan. For this study, only angles about the flex/extension axis were reported since movement occurred primarily in the sagittal plane. Joint angles during the trials were represented as a deviation from the quiet standing calibration trial. Lumbopelvic angles were expressed as a percentage of maximum flexion or extension obtained during the posture normalization trials, depending on the direction of the angle.

Raw force plate data were low pass filtered at 10 Hz to remove high frequency noise using a Butterworth filter (2nd order, dual pass) and expressed as a percentage of body weight (%BW), which was measured during the quiet standing calibration trial. Each direction of force (vertical, A/P), and M/L) was computed separately.

An average from the last 30 seconds of each one minute trial was calculated for all dependent variables and used for the statistical analysis. To assess whether the lumbopelvic and pelvic angles between males and females were equal to each other, a statistical approach employing the normal-based parametric bootstrap MaxT (Westfall 1997) appropriately modified for high-dimensional data (Konietschke, Gel, and Brunner 2014) was performed using R3.3.0 (R Development Core Team 2018). The mean values from each trunk-thigh angle (total of 19 data points) were compared between males and females. To assess where the lumbopelvic and pelvic angles deviated significantly from either standing or sitting postures, a statistical approach employing the normal-based parametric bootstrap MaxT for high-dimensional data with Dunnett's contrast using R3.3.0 (R Development Core Team 2018) was performed. Equality of means was assessed for the following scenarios: i) between sitting (90° trunk-thigh angle) and

the remaining posture trials (95 to 180° trunk-thigh angles), and ii) between standing (180° trunk-thigh angle) and the remaining posture trials (90 to 175° trunk-thigh angles). Using the lumbopelvic angle results, the trials where significant differences occurred were determined to be the breakpoints from sitting and standing, and those postures in between were designated as perching. The same procedure was repeated for each sex. For both MaxT tests, the adjusted p-value of 0.05 was used to indicate significance. Using the classification from each sex, an average of each phase (sitting, perching, and standing) was calculated and one-way repeated measures ANOVAs were performed (SPSS Inc., Chicago, IL) to determine the effect of postural phases on EMG and force data ($p < 0.05$).

3. Results

3.1 Protocol Considerations

The direction of seat movement was consistent (upwards from sitting to standing) across participants since a pilot study ($n = 3$) demonstrated that there were minimal differences in the root mean square error (RMSE) of lumbopelvic and pelvic angles (3.74° and 4.38°, respectively) between trials starting from sitting and from standing.

The two controlled variables in this study were trunk-thigh angle and shank angle. The RMSE between the actual (measured during the experiment) and target (calculated based on the ellipse equation) trunk-thigh angles were 9.75° and 6.81° for males and females, respectively (Figure 2). The greatest deviation occurred at 150° trunk-thigh angle, where the difference between the actual and target angles was approximately 13°. It should be noted that the trunk-thigh angles in the results represent the target angles. Shank angle was maintained within $\pm 1.1^\circ$

on average throughout the trials. Postural symmetry was maintained for the left and right hips and knees as they differed by only 2.5% (RMSE 2.13°) and 4% (RMSE 3.23°), respectively.

The seat location was standardized to the length of the thigh, and therefore varied between sex (Figure 3), while the seat tilt angle tracked the trunk-thigh angle closely and was similar between males and females (Figure 4), indicating that the seat interface between participants remained consistent.

Figure 2: Comparison of actual (measured) and target (calculated) trunk-thigh angles from sitting (90° trunk-thigh angle) to standing (180° trunk-thigh angle) postures. The lightest gray line represents the ideal scenario where actual and target trunk-thigh angles matched. The dotted lines indicate where the actual 135° trunk-thigh angle occurred.

Figure 3: Vertical and horizontal location of the seat pan (defined as the location of the seat pan hinge, which was located at the centre of the seat pan to allow for sagittal rotation) from sitting (90° trunk-thigh angle) to standing (180° trunk-thigh angle) postures. The differences in seat pan location occurred due to standardizing trunk-thigh angles.

Figure 4: Seat pan tilt angle from sitting (90° trunk-thigh angle) to standing (180° trunk-thigh angle) postures.

3.2 Sex Difference in Sitting, Hybrid Postures, and Standing

There were differences between males and females in lumbopelvic ($p = 0.0015$; Cohen's $d = -1.012$) and pelvic ($p = 0.0116$; Cohen's $d = 0.941$) angles during sitting (90° trunk-thigh

angle). Males showed a greater lumbopelvic flexion and posterior pelvic tilt at 90° trunk-thigh angle compared to females. The remaining postures did not show any sex difference.

3.3 Kinematic Results from Sitting to Standing

Standing lumbopelvic and pelvic angles were different from all other trunk-thigh angles for both males and females ($p < 0.0001$), as participants maintained distinctively more extended lumbopelvic and anteriorly tilted pelvic postures in standing. Conversely, lumbopelvic angle did not significantly deviate from sitting until a trunk-thigh angle of 145° ($p < 0.05$; range: $< 0.0001 - 0.024$; Cohen's d range = $1.623 - 4.884$) and 160° ($p < 0.05$; range: $< 0.0001 - 0.0456$; Cohen's d range = $1.329 - 5.019$) for males and females, respectively (Figure 5 top). As a result, the perching phase was determined to be from 145 to 175° for males and from 160 to 175° for females. Pelvic angle was different from sitting at 180° trunk-thigh angle for both males ($p = 0.0031$; Cohen's $d = -2.211$) and females ($p = 0.0350$; Cohen's $d = -1.416$) (Figure 5 bottom). Qualitatively, both lumbopelvic and pelvic angles changed non-linearly from sitting to standing. This trend was particularly noticeable in females as their lumbopelvic angle became more flexed and pelvic angle became more posteriorly tilted initially from 90° to approximately 115° trunk-thigh angles.

Figure 5: Lumbopelvic (top) and pelvic (bottom) angles from sitting (90° trunk-thigh angle) to standing (180° trunk-thigh angle) postures. Positive values indicate flexion and posterior pelvic tilt, respectively. The dagger (†) symbol for males and asterisk (*) symbol for females denote

significant difference from the angles in sitting (dotted square; $p < 0.05$). The angles in standing were significantly different from the rest of hybrid sit-stand postures (not shown on the graph).

3.4 Muscle Activity in Sitting, Perching, and Standing

The statistical results from muscle activity and external forces are based on the comparison of mean value in each phase (sitting = M: 90–140°, F: 90–155°; perching = M: 145–175°, F: 160–175°; standing = M and F: 180°). The level of muscle activity, on average, was low ($< 10\%$ MVC) in all muscles throughout the trials (Table 1 and 2; Figure 6). Males exhibited no difference between phases for both trunk extensors (TES $p = 0.247$; LES $p = 0.319$), while significantly lower VMO activity was seen in standing compared to perching or sitting ($p = 0.005$). Females showed significant differences in all muscles: TES was higher in sitting compared to perching or standing ($p < 0.005$), LES was higher in standing compared to perching ($p = 0.015$), and VMO was lower in standing compared to perching or sitting ($p = 0.003$).

Table 1: Mean (SD) EMG and force values in each phase for males.

Table 2: Mean (SD) EMG and force values in each phase for females.

Figure 6: Male and female VMO (top) and female ES (bottom) muscle activities from sitting (90° trunk-thigh angle) to standing (180° trunk-thigh angle) postures. Different alphabets (A and

B) above the below the curves denote significant difference between phases for each sex ($p < 0.05$). Statistical results for males are in black (top) and females are in grey (bottom).

3.5 Ground Reaction Force in Sitting, Perching, and Standing

In terms of ground reaction force, all phases were different from each other in both sexes in the A/P direction ($p < 0.001$), in which perching had the highest magnitude requiring approximately 15% BW and standing had the lowest magnitude requiring approximately 2% BW to maintain posture (Table 1 and 2; Figure 7 top). Both vertical and M/L forces increased as participants moved from sitting to standing (Figure 7 bottom). All pairwise comparisons were also different in both sexes ($p < 0.001$), except for the comparison between perching and standing in M/L forces in females.

Figure 7: A/P (top) and M/L (bottom) forces from sitting (90° trunk-thigh angle) to standing (180° trunk-thigh angle) postures. Positive values indicate anterior and lateral direction of force, respectively. Different alphabets (A, B, and C) above the below the curves denote significant difference between phases for each sex ($p < 0.05$). Statistical results for males are in black (top) and females are in grey (bottom).

4. Discussion

Standing lumbopelvic and pelvic angles were different from all other trunk-thigh angle trials for both males and females, suggesting that standing is a distinct posture and lumbar

lordosis is difficult to maintain once perching is initiated. Lumbopelvic angle was different from sitting starting from 145 and 160° trunk-thigh angles for males and females, respectively, while pelvic angle remained similar throughout the range of hybrid sit-stand postures. While perching improved lumbopelvic angles compared to sitting, there was a large increase in ground reaction A/P shear force compared to sitting, a large increase in VMO activity compared to standing, and little to no change in trunk extensor activities compared to both sitting and standing. Taken together, these results demonstrate the potential of perching as an alternative posture to sitting and standing, especially if seat and chair design can be improved to minimize lower limb demands.

To the best of the authors' knowledge, this is the first study to characterize lumbopelvic and pelvic angles throughout the entire range from sitting to upright standing. Angles measured in this study were within the range of previously reported angles: lumbopelvic and pelvic angles in sitting were $74.6 \pm 19.1\%$ ROM and $22.4 \pm 10.2^\circ$ for males and $59.0 \pm 10.6\%$ ROM and $14.9 \pm 4.5^\circ$ for females, while previous studies reported values within 50-70% ROM (O'Sullivan et al. 2012; Dunk and Callaghan 2005; Beach et al. 2005; De Carvalho and Callaghan 2011) and $12-30^\circ$ (Dunk and Callaghan 2005; De Carvalho and Callaghan 2011), respectively. Despite linear increases in both target trunk-thigh angles and mass distribution from the seat pan to the feet, changes in lumbopelvic and pelvic angles from sitting to standing followed higher order polynomial trends. This trend may be partially attributable to the fact that pelvic angle is difficult to systematically change when the pelvis is in contact with any supporting surface. Since the majority of spine flexion comes from the lowest spine segment (i.e., L5/S1 (Dunk et al. 2009; De Carvalho 2015)), lumbopelvic angles would also be difficult to systematically change, as was shown in the results that lumbopelvic angles were similar to sitting in more than half of the

hybrid sit-stand postures. These results are in agreement with a recent radiographic study, in which the lumbar spine was flexed to approximately 70% ROM regardless of the adjustment of various chair features (De Carvalho 2015).

The breakpoint between sitting and perching was different between sexes, which can be attributed to the differences in sitting lumbopelvic angles. Females tend to maintain a more upright posture (less lumbar flexion and more anterior pelvic tilt) compared to males in sitting as previously reported (Dunk and Callaghan 2005; Beach et al. 2005; Beach et al. 2008), resulting in a wider range of trunk-thigh angle postures that were similar to sitting. In other words, it requires a greater trunk-thigh angle (i.e., a higher stool) for females to achieve significantly different lumbopelvic angles compared to sitting. This result should not be mistaken for females maintaining less upright postures during hybrid sit-stand postures: their lumbopelvic and pelvic angles were in fact more extended and anteriorly tilted (closer to upright standing), respectively, than males from 90° trunk-thigh angle until approximately 155° trunk-thigh angle. Differences in sex may be due to the location and magnitude of the centre of mass with respect to the chair hinge. A study comparing sex and different types of chairs found that males tend to shift their centre of mass closer to the base of the chair when sitting on a pivot chair, compared to a standard office chair, in order to maintain balance (Dunk and Callaghan 2005). Perhaps for males, it is easier to maintain equilibrium between the mass of the lower limbs and upper body when the upper body is in a flexed posture in sitting. Further investigation is required to understand this complex relationship between posture and centre of mass, centre of pressure, chair hinge location, and the distribution of stress on the seat pan.

It has been shown that sitting with a larger trunk-thigh angle (i.e., > 90°) results in the lumbar angle closer to its neutral orientation (Keegan 1953; Alexander et al. 2007; Hirasawa et

al. 2007) and a large reduction in spinal loads (Rohlmann et al. 2011). The trunk-thigh angle that was recommended in the literature is approximately 135° (Keegan 1953; Rohlmann et al. 2011). This study demonstrated that both lumbopelvic and pelvic angles at a 135° target trunk-thigh angle were not statistically different from their respective angles in sitting. However, this result should be taken with caution as 1) the seating device used in this study did not have a backrest and 2) the actual trunk-thigh angle in this trial was smaller than the target trunk-thigh angle (Figure 2). Using a backrest and having the ability to adjust the backrest showed a reduction in discomfort (Groenesteijn et al. 2009; Vergara and Page 2000; Gregory, Dunk, and Callaghan 2006), while other studies comparing chairs with and without a backrest showed no change in discomfort (Curran et al. 2014) and no difference in LES activity level (Curran et al. 2014) and trunk co-contraction (Gregory, Dunk, and Callaghan 2006). By not having the backrest in this study, the responses seen in this study can be confidently attributed to the changes in trunk-thigh angles. In this study, the actual trunk-thigh angle was not used to standardize each trial in order to avoid giving instructions to the participants to move their trunk, which in turn could affect the postural response. The actual 135° trunk-thigh angle occurred at around 140 to 150° target trunk-thigh angle trials (Figure 2), which, interestingly, is where lumbopelvic angle became significantly different from sitting for females. It is difficult to conclude whether the 135° trunk-thigh angle is optimal for maintaining the neutral spine posture but it is within the range where the lumbopelvic angle deviates from that of sitting.

While perching tended to improve lumbopelvic angle, there was a substantial increase in lower limb demand as shown by an increase in A/P shear force and VMO activity. Peak A/P shear approached 20% BW at 155° trunk-thigh angle and peak VMO approached 10% MVC at 145° trunk-thigh angle. The 20% BW A/P shear force approximates the magnitude experienced

during walking heel strike and toe off (Kirtley 2003), and the VMO activity of 10% MVC is greater than the average activity level experienced during walking (Arsenault, Winter, and Mrteniuk 1986). This increase in lower limb demand occurs as a result of having to resist the upper body mass from sliding forward and down from the seat pan as the seat pan used in this study was flat and had minimal surface friction. If this magnitude of loading is maintained for a prolonged period of time, it is possible to experience an increase in discomfort (Mak, Zhang, and Tam 2010) and muscle overload due to the static and prolonged (i.e., without breaks) nature of the exposure (Jonsson 1978). One way to reduce lower limb demands is by implementing an appropriate foot support system (Antle, Vezina, and Cote 2015); for example a foot rest that can align the ankle in a neutral posture and the shank angle parallel to the direction of the resultant ground reaction force. However, this additional equipment may encumber the individual's workspace while reducing workstation adjustability. Therefore, aiming design improvements at optimizing the seat pan interface for trunk-thigh angles between 115 and 170° (a range where VMO >5% MVC) may make perching postures far more achievable.

These results should be considered with several limitations in mind. In addition to the aforementioned lack of backrest, the seat interface used in this study was made of MDF, which was flat and had a lower coefficient of friction compared to traditional chairs with soft materials covering a contoured seat pan. This interface eliminated any confounding factors associated with the choice of cushion (i.e., shape, thickness, and stiffness) influencing the pelvic and lumbopelvic angles. Secondly, participants were instructed to have their feet in the same orientation and keep their sagittal shank angle within $\pm 2^\circ$ for all 19 trials. These constraints were necessary to alter trunk-thigh angles in a reproducible manner and without adopting a different strategy to maintain perching postures (e.g., locking knees in extension and using friction of the

floor to counteract forces). Lastly, it should be acknowledged that the order of the trunk-thigh angles tested was not randomized. With the current custom-made seating device, the between-participant repeatability was higher when the trunk-thigh angle was changed sequentially. In order to minimize carry-over effects, a two-minute standing rest was given between each trial.

5. Conclusions

Characterizing lumbopelvic angle from sitting to standing indicated that significantly less flexion occurs at an actual trunk-thigh angle of 135° or greater. On the other hand, the pelvic angle was less posteriorly tilted only in the standing trial compared to sitting. Taking both EMG and force data into consideration, the notable differences between the perching and sitting/standing phases were in the level of leg effort required to remain upright. There is no ‘ideal’ seat position that includes all of the benefits from sitting, perching, and standing; however, future chair designs should be focused on trunk-thigh angles between 115° and 170° , while keeping these two key points in mind: 1) the seat interface should focus on pelvis alignment to improve lumbar angles and to offload some of the lower limb demands, and 2) the design of the foot support (if at all) should focus on redirecting force from shear to compression with respect to the lower limb joints while reducing user and workstation encumbrance.

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Disclosure statement

No conflict of interest was reported by the authors.

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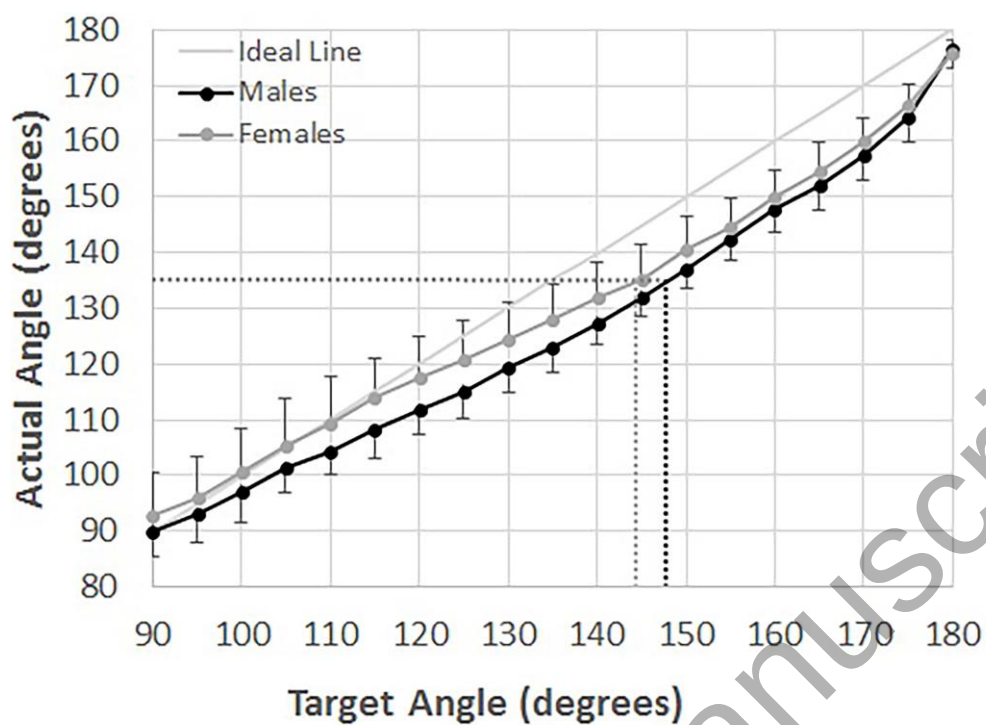
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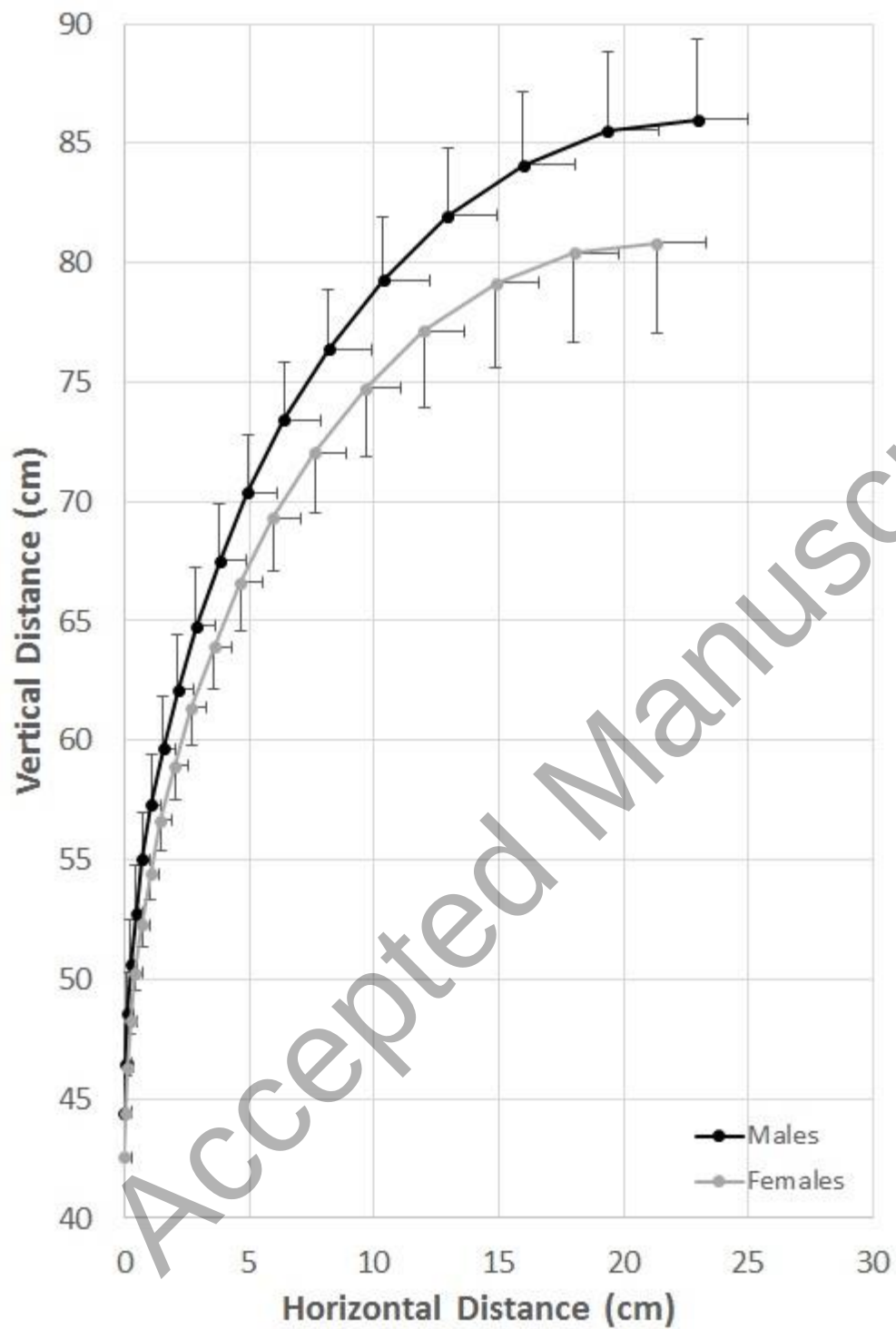
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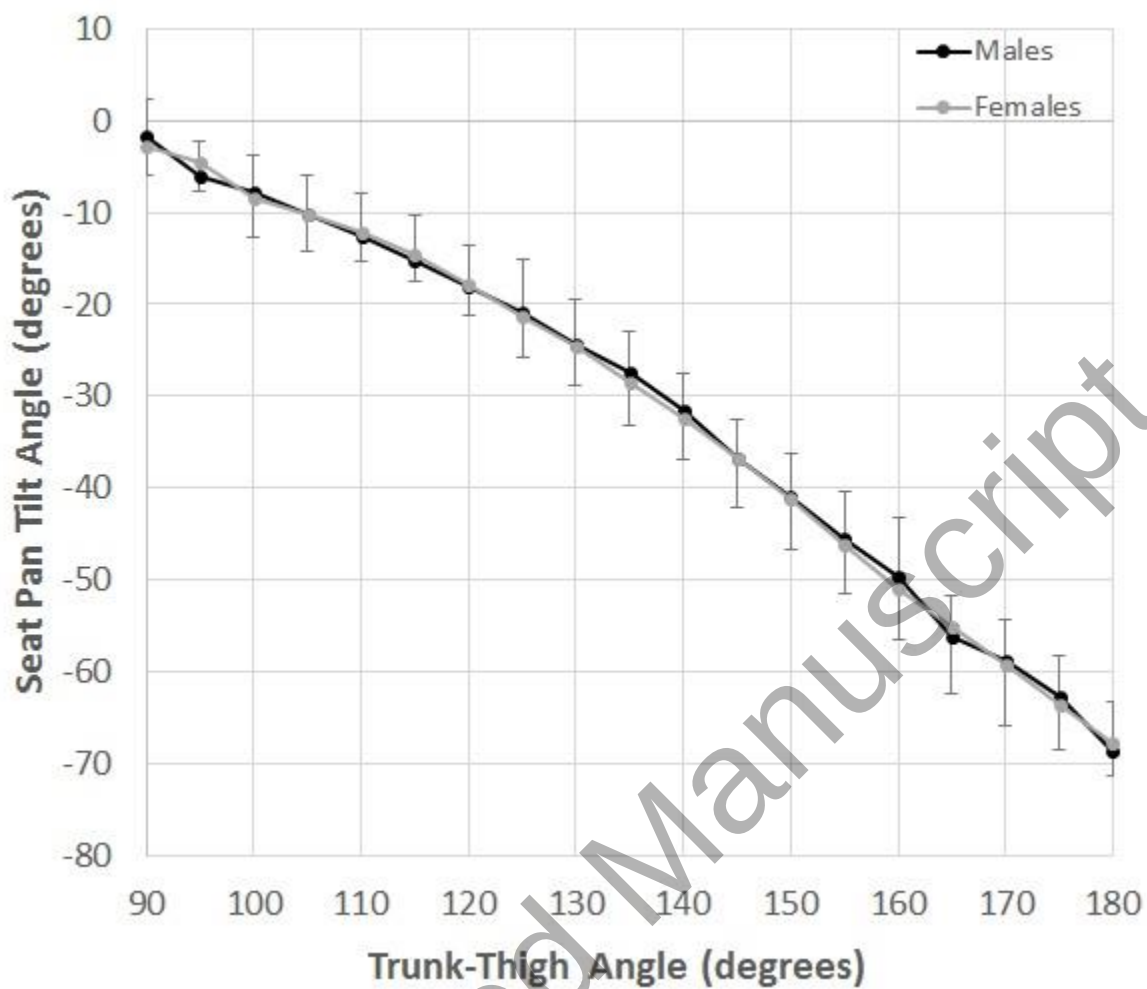
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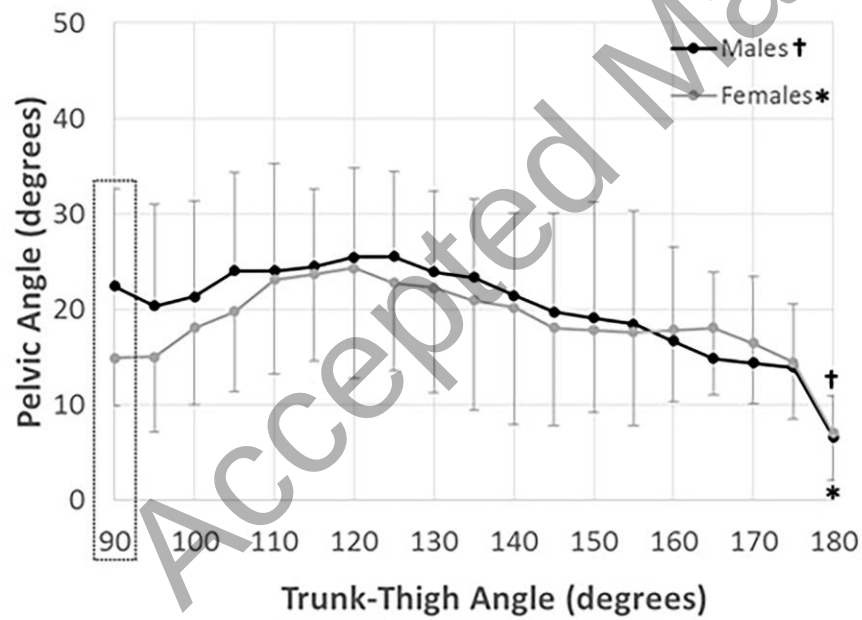
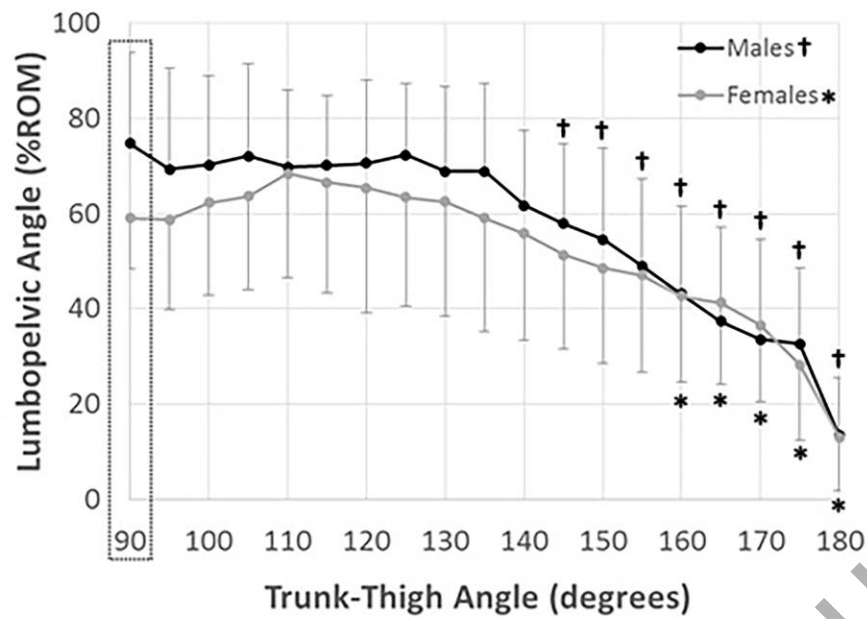
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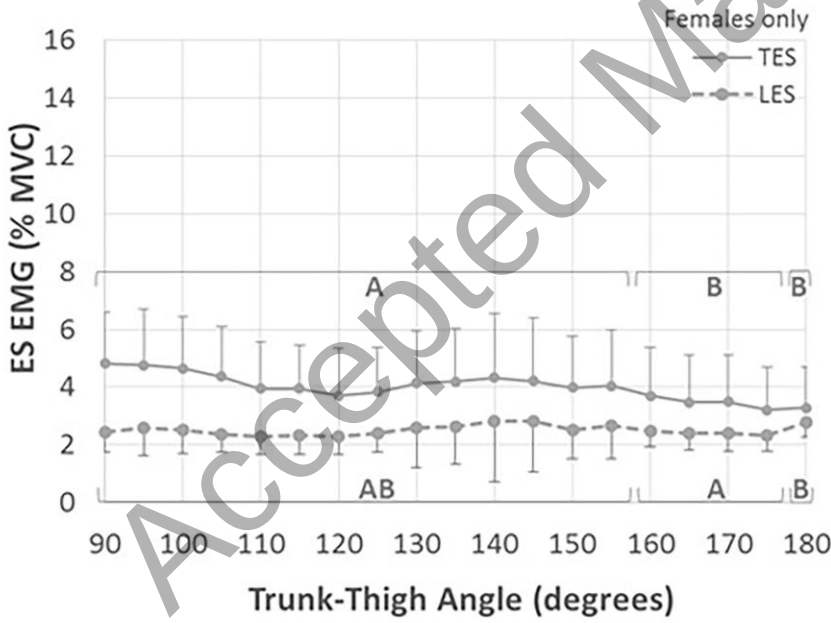
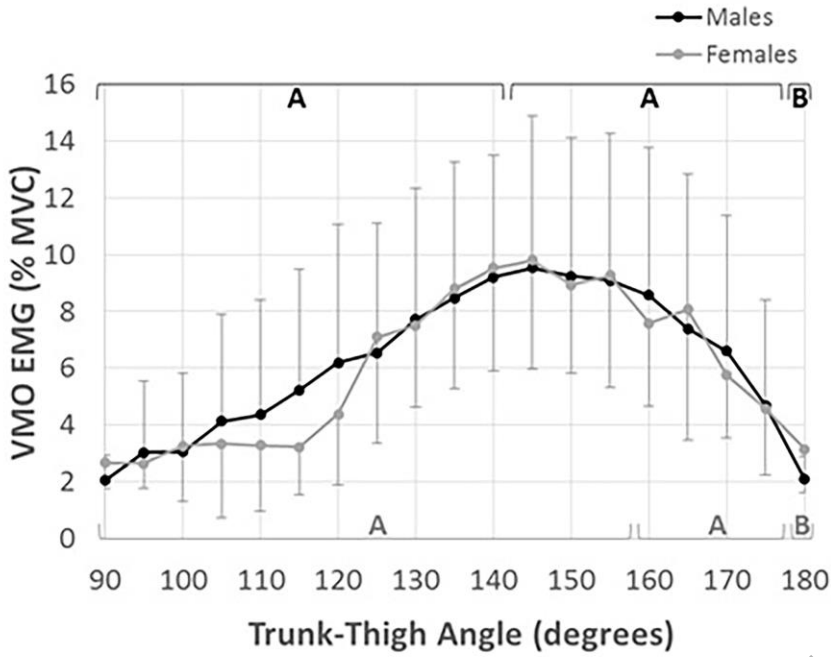












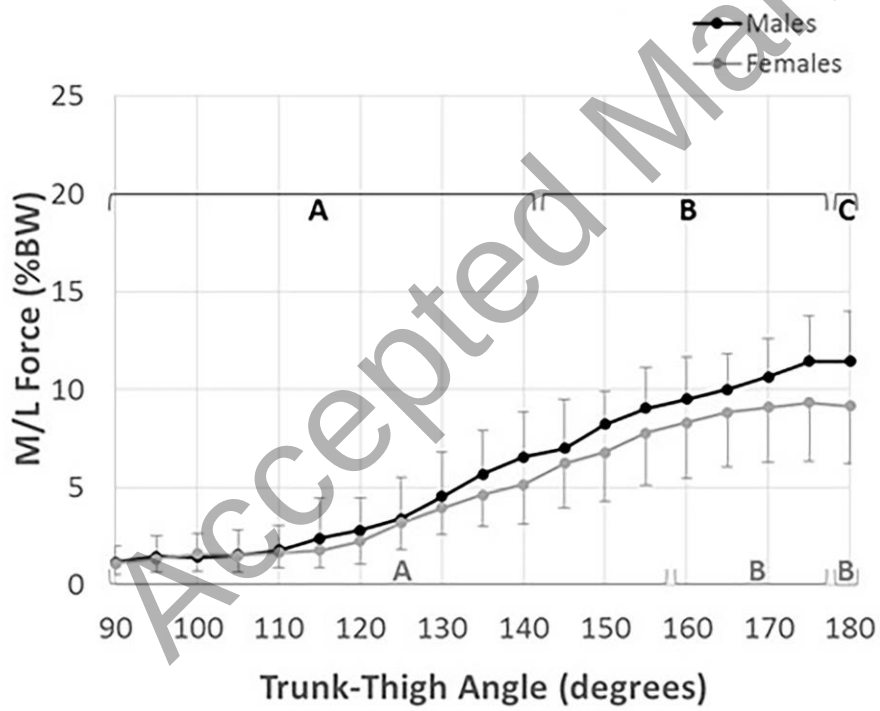
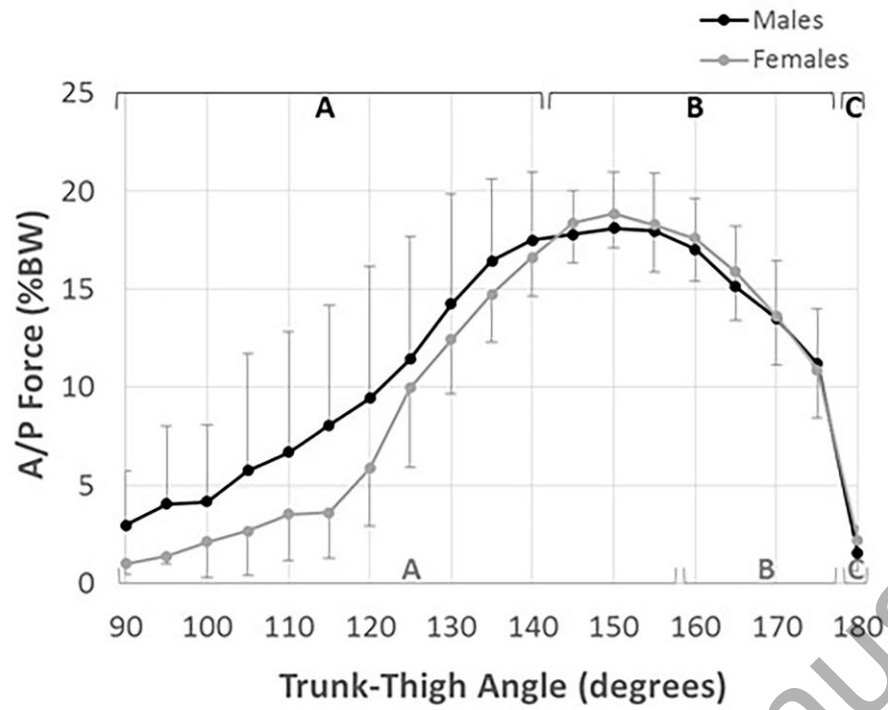


Table 1: Mean (SD) EMG and force values in each phase for males.

		Sitting	Perching	Standing
EMG (%MVC)	VMO	5.45 (3.46)	7.87 (4.54)	2.10 (0.78)
	TES	3.37 (1.40)	3.37 (1.61)	3.03 (1.62)
	LES	2.62 (1.83)	2.85 (1.79)	3.28 (1.99)
Force (%BW)	Vertical	37.66 (8.30)	82.07 (3.07)	100.00 (0.00)
	A/P	9.16 (4.56)	15.82 (2.45)	1.54 (1.41)
	M/L	2.98 (1.48)	9.41 (1.87)	11.45 (2.54)

Table 2: Mean (SD) EMG and force values in each phase for females.

		Sitting	Perching	Standing
EMG (%MVC)	VMO	5.98 (2.32)	6.50 (2.82)	3.14 (1.53)
	TES	4.21 (1.61)	3.47 (1.56)	3.29 (1.40)
	LES	2.52 (0.96)	2.40 (0.56)	2.77 (0.49)
Force (%BW)	Vertical	41.95 (4.08)	89.01 (2.60)	100.00 (0.00)
	A/P	9.26 (1.10)	14.50 (2.12)	2.21 (1.58)
	M/L	3.50 (1.20)	8.89 (2.83)	9.16 (2.92)